

## 3.4 Differentiation

### Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

Differentiate composite trig, exponential and logarithmic functions with chain, product and quotient rules. Implicit/parametric second derivatives require care:  $\frac{d^2y}{dx^2} = \frac{d}{dt}(\frac{dy}{dx}) / (\frac{dx}{dt})$ . Identify stationary points and inflection using derivatives and sign tests.

#### Worked example: follow the reasoning

If  $y = \ln(2x+1)$ ,  $y' = 2/(2x+1)$  on  $x > -1/2$ .

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Differentiate  $y = x e^{(2x)}$  and find  $y'$  at  $x=0$ .

[3]

Final answer: \_\_\_\_\_

## Answers and explanation

### Question 1 [3 marks]

Product and chain:  $y' = e^{(2x)} + 2xe^{(2x)} = e^{(2x)}(1+2x)$  [2]; at  $x=0$ , 1 [1].

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.